



Planta y Diviértete

Learning From Plants to Design Regenerative Systems

for Future Communities

Independent Research Proposal

Author: Ana Lucía Ochoa López

Bachelor of Science in Environmental and Energy Systems Engineering

Lucerne University of Applied Sciences and Arts (HSLU)

May 29, 2026.

HSLU Hochschule
Luzern

Executive Summary

Planta y Diviértete is an interdisciplinary research initiative that investigates how biological principles observed in plants and living ecosystems may inspire future regenerative systems, products, and engineering approaches.

The project originates from more than six years of plant cultivation experience across Mexico, Canada, and Switzerland, combined with academic studies in Environmental and Energy Systems Engineering. Through long-term observation of plant adaptation, resource management, resilience, and ecological interactions, the project seeks to explore how natural systems can inform future engineering solutions.

Rather than focusing solely on sustainability, the research adopts a regenerative perspective, investigating how future systems may actively contribute to ecosystem restoration, environmental resilience, and human wellbeing.

The long-term objective is to establish a research platform connecting biomimicry, ecological engineering, regenerative design, and environmental innovation.

1. Background and Context

Modern technological systems have generated remarkable progress but often operate through linear processes that consume resources and generate waste.

Natural systems function differently.

Plants continuously adapt to changing conditions while contributing positively to the ecosystems around them. They capture solar energy, regulate environmental conditions, support biodiversity, recycle nutrients, and participate in regenerative cycles.

These observations raise an important question:

Can future engineering systems learn from the strategies already developed by living systems?

This research seeks to investigate that possibility.

2. Problem Statement

Most human systems are designed to reduce negative environmental impacts rather than create positive ecological outcomes.

While sustainability aims to minimize harm, future environmental challenges may require approaches capable of restoring ecosystems and strengthening ecological relationships.

There remains a need to investigate how biological systems, particularly plants, can inspire regenerative approaches to engineering, design, and environmental problem-solving.

3. Research Question

Main Research Question

How can humans learn from plants and living ecosystems to design regenerative systems that contribute positively to both environmental and human communities?

Supporting Questions

- How do plants manage energy and resources?
 - How do plants adapt to environmental change?
 - Which regenerative principles exist within ecosystems?
 - How can engineering systems learn from biological cycles?
 - How can future technologies contribute to ecological restoration?
-

4. Research Objectives

Main Objective

To investigate how biological principles observed in plants may inspire regenerative systems, products, and engineering approaches.

Specific Objectives

- Study plant adaptation mechanisms.
 - Investigate ecological resilience strategies.
 - Explore biomimicry and regenerative design principles.
 - Develop conceptual frameworks connecting biology and engineering.
 - Identify future opportunities for nature-inspired innovation.
-

5. Conceptual Framework

This research combines knowledge from:

- Biomimicry
- Regenerative Design
- Systems Thinking
- Ecological Engineering
- Nature-Based Solutions
- Environmental Engineering

Together, these fields provide a foundation for understanding how biological systems may inspire future technologies and environmental solutions.

6. Methodology

The project will be developed through:

Literature Review

Scientific review of:

- Plant physiology
- Biomimicry
- Regenerative design
- Ecological engineering
- Nature-based solutions

Observation-Based Research

Long-term plant observation and documentation.

Systems Analysis

Identification of:

- Inputs
- Outputs
- Feedback loops
- Resource flows
- Ecological interactions

within living systems.

Concept Development

Development of conceptual models and future application scenarios.

7. Expected Outcomes

The project aims to produce:

- Research documentation
 - Conceptual frameworks
 - Plant-inspired system models
 - Future project proposals
 - Educational resources
 - Potential interdisciplinary collaborations
-

8. Long-Term Vision

The long-term vision of Planta y Diviértete is to build a multidisciplinary platform where engineers, ecologists, designers, educators, and communities collaborate to learn from living systems.

The objective is not simply to create products inspired by nature, but to investigate how future systems can actively regenerate ecosystems while supporting human wellbeing.

Ultimately, the project seeks to contribute to a future where engineering learns from nature not only as a resource, but as a source of knowledge, adaptation, resilience, and regeneration.

References

Biomimicry Institute

AskNature

United Nations Environment Programme (UNEP)

International Union for Conservation of Nature (IUCN)

Ecological Engineering Literature

Regenerative Design Literature

Nature-Based Solutions Literature

Environmental and Energy Systems Engineering Studies

Personal Plant Observation Records (2020–Present)